Part 1 Write-Up

Data Understanding

Ten files were provided: nine data tables (Customers, Products, Orders, Stores, Returns, Employees, Suppliers, Inventory, Order_Items) and one Data Dictionary.

Dataset	Represents
Customers	Customer master data (demographics, location, segment, registration)
Products	Product catalog (name, category, brand, cost price, selling price)
Orders	Sales transactions (who bought what, from which store, when, at what price)
Stores	Store master data (name, region, state, opening date)
Returns	Product returns and refunds linked to orders
Employees	Staff master data (role, store assignment, salary)
Suppliers	Supplier master data (contact details, country)
Inventory	Stock levels per product per store, with reorder thresholds
Order_Items	Line-item detail for orders
Data_Dictionary	Column definitions for 5 tables (Customers, Orders, Products, Stores, Returns)

Primary keys: A Primary Key (PK) uniquely identifies every row within a dataset table without null or duplicate values. In this worksheet, each table uses its own ID (Customer_ID, Product_ID, Order_ID, Store_ID, Return_ID, Employee_ID, Supplier_ID, Inventory_ID, Order_Item_ID).

Foreign keys: A Foreign Key (FK) is a column that creates a relational bridge between tables by referencing the primary key of a parent entity table. Orders references Customer_ID, Product_ID, Store_ID. Returns references Order_ID. Employees references Store_ID. Inventory references Product_ID and Store_ID. Order_Items references Order_ID and Product_ID.

Relationships:

